SUPERVISED MACHINE LEARNING

CUSTOMER CHURN PREDICTION

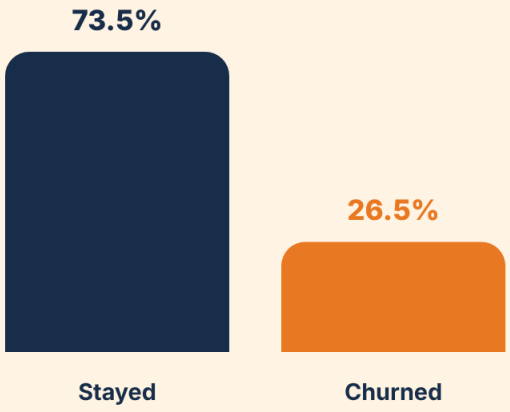
Step 1: Problem Framing & Theory Notes

Frame. Measure. Predict.

A clear read of the theory behind the project: what churn is, how prediction errors are costed, how class imbalance is handled and how the four algorithms work.

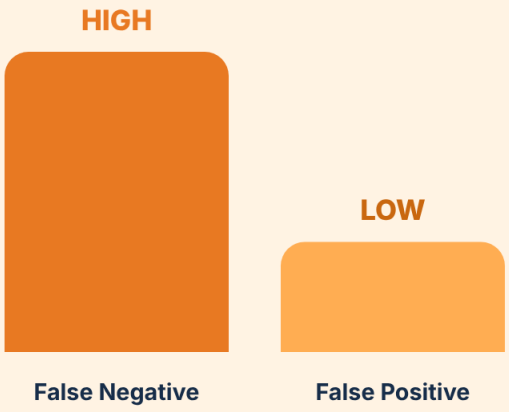
Class balance

share of customers in the Telco dataset



Cost of a wrong prediction

business impact, churn context



26.5%
customers churned

4
algorithms compared

5
theory questions answered

01 Frame
Churn, CAC vs CLV

02 Measure
Confusion matrix, precision and recall

03 Balance
Class imbalance, SMOTE, four algorithms

Q Answer the five theory questions below.

Every answer is written for the Telco churn dataset, where about 26.5% of customers churned and 73.5% stayed.

1 What is Customer Churn? Why is predicting churn valuable for a business like Jio or Airtel? (Mention CAC vs CLV in your answer.)

Churn	Churn means a paying customer leaves the service (stops the subscription).
CAC	Getting a new customer (CAC - Customer Acquisition Cost) is much more expensive than keeping an old one.
CLV	Every churned customer also takes away his future revenue (CLV - Customer Lifetime Value).
Why predict	If Jio/Airtel can predict who will leave, the retention team can give offers in time and protect the revenue.

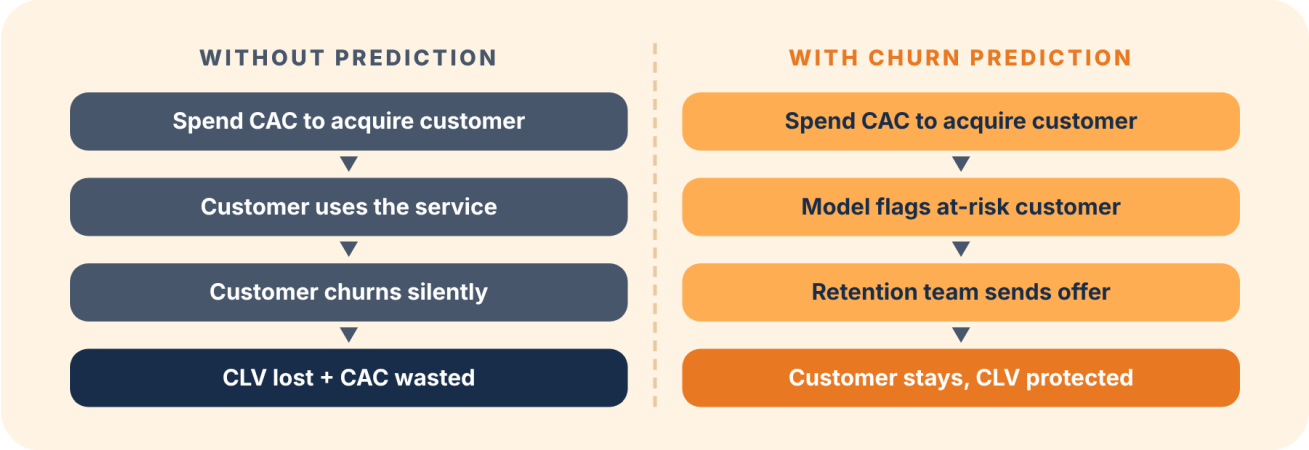


Figure 1: How churn prediction protects CAC and CLV

2 Define the four cells of a Confusion Matrix (TP, TN, FP, FN). In the churn context, what is the business cost of a False Negative? What about a False Positive?

TP and TN	TP: customer actually churned and model predicted churn. TN: customer actually stayed and model predicted stay.
FP and FN	FP: customer actually stayed but model predicted churn. FN: customer actually churned but model predicted stay.
Cost of FN	Cost of FN is high: we lose the customer and his CLV, and we never tried to save him.
Cost of FP	Cost of FP is low: we only waste a small offer or a call on a customer who would stay anyway.

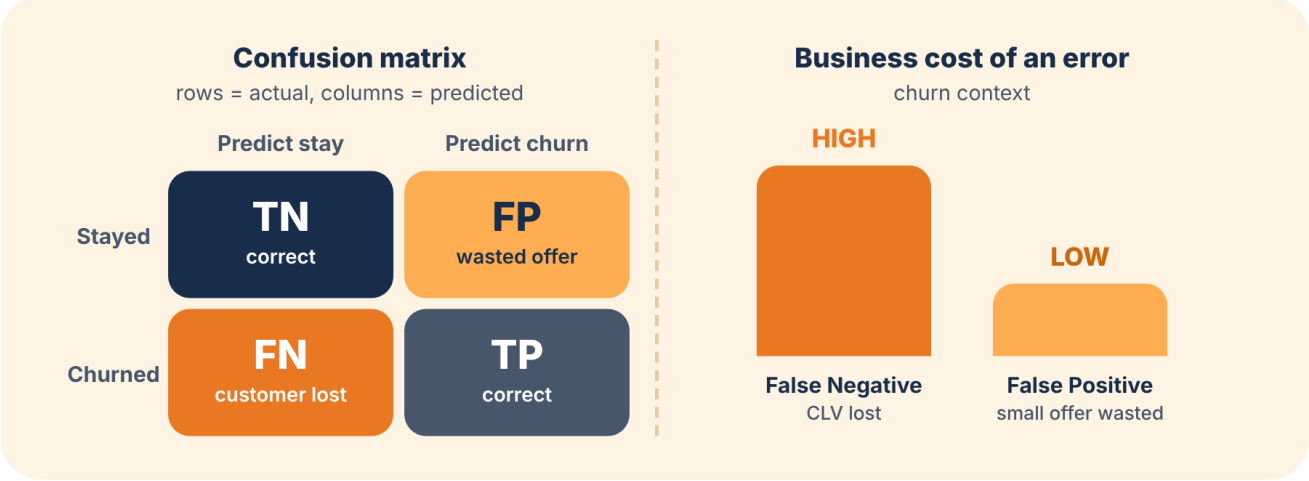


Figure 2: Confusion matrix (left) and the business cost of each error (right)

3 What is Class Imbalance? Does the Telco Churn dataset have it? How does SMOTE address it?

Class imbalance	Class imbalance means one class has far less data than the other class, so the model becomes biased to the majority class.
Telco data	Yes, Telco data is imbalanced - about 26.5% customers churned and 73.5% stayed (confirmed in Step 2).
SMOTE	SMOTE creates new synthetic samples of the minority class using its nearest neighbours, so both classes become equal in training data.

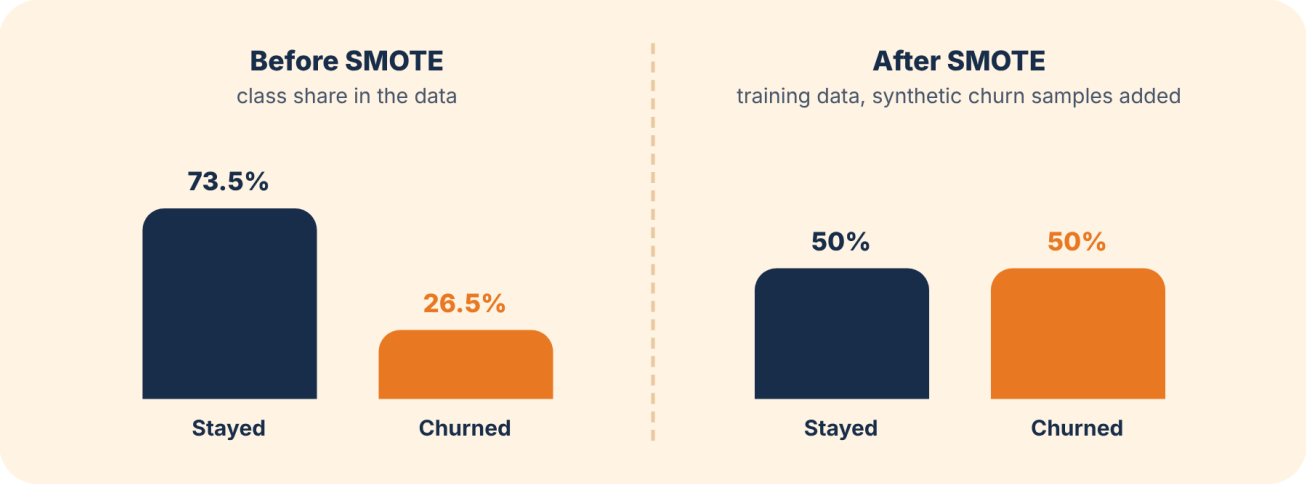


Figure 3: Class balance in the Telco data before and after SMOTE

4 Briefly explain how each of the 4 algorithms you will use works: KNN, Naive Bayes, SVM, Decision Tree (2-3 lines each).

Algorithm	How it works	Key idea
KNN	It stores all the training data. For a new customer it finds the k nearest customers and takes the majority class among them.	Majority vote
Naive Bayes	It uses Bayes theorem to find the probability of each class given the features, and it assumes all features are independent.	Bayes rule
SVM	It finds the best boundary (hyperplane) with the maximum margin between the two classes. With the RBF kernel it can make a non-linear boundary.	Maximum margin
Decision Tree	It keeps splitting the data with yes/no questions on features (like Contract = Month-to-month?) and each leaf gives the final class.	Yes/no splits

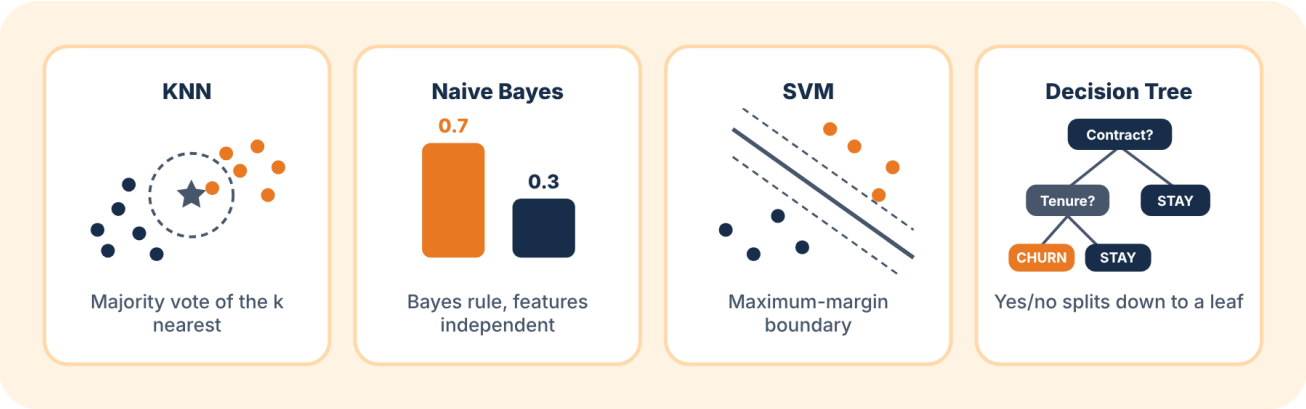


Figure 4: How the four algorithms make a prediction

5 When would you choose Precision over Recall as your primary metric? Give a churn-specific answer.

Precision first	Precision is more important when the retention action is costly or the calling capacity is limited.
Example	If the team can call only ~500 customers a day or gives a big discount, wrongly flagged loyal customers waste money and call slots.

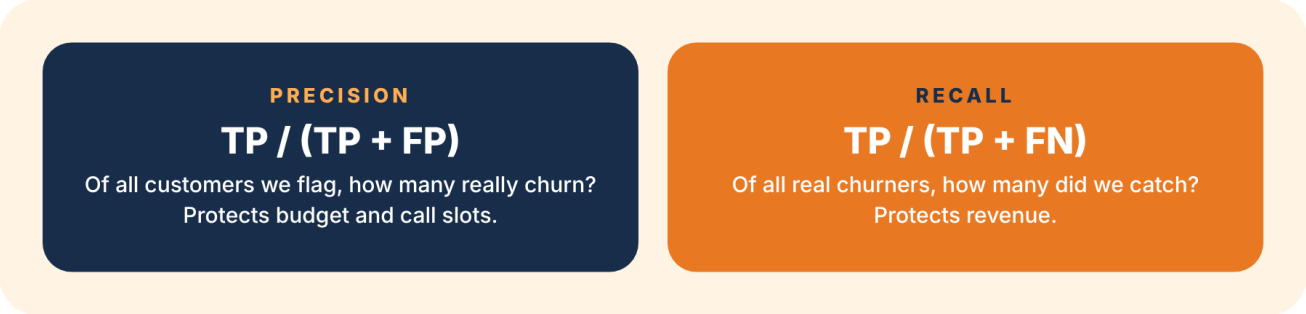


Figure 5: Precision and recall: what each one protects

6 Key takeaways for the churn model

RECALL | PROTECT REVENUE	PRECISION | PROTECT BUDGET
Catch every likely churner	**Flag only likely churners**
A missed churner (FN) costs the customer and the CLV.	Best when offers are costly or call capacity is limited.

Data	About 26.5% of Telco customers churned and 73.5% stayed, so SMOTE balances the training data.
Models	KNN, Naive Bayes, SVM and Decision Tree are compared on the same churn data.
Metrics	Recall protects revenue by catching churners. Precision protects budget when offers or calls are limited.

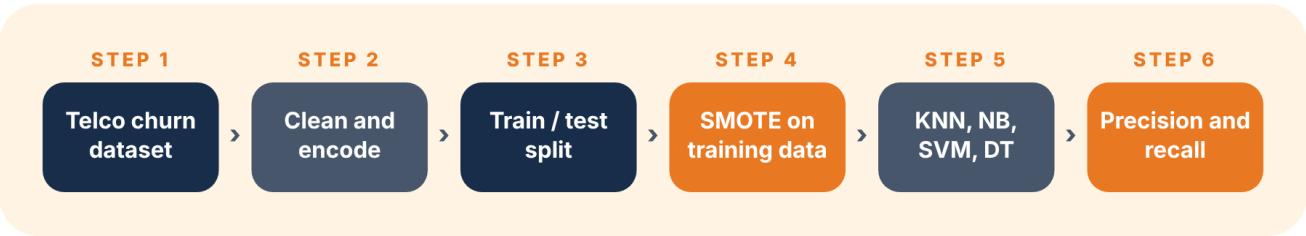


Figure 6: Where this theory fits in the project workflow

7 Quick glossary

Churn	A paying customer leaves the service (stops the subscription).
CAC	Customer Acquisition Cost: what it costs to win a new customer.
CLV	Customer Lifetime Value: the future revenue a customer would bring.
TP / TN	Correct predictions: churn predicted and churned, or stay predicted and stayed.
FP	Churn predicted but the customer stayed. Low cost: a small offer or call is wasted.
FN	Stay predicted but the customer churned. High cost: the customer and CLV are lost.
Class imbalance	One class has far less data than the other, so the model leans to the majority.
SMOTE	Creates synthetic minority-class samples from nearest neighbours to balance training data.
Precision	Of all customers flagged as churners, the share who really churn: TP / (TP + FP).
Recall	Of all real churners, the share the model catches: TP / (TP + FN).